⑤ KL, MLE, VLB/ELBO:

想要 minimize $KL(q(x)\|P_\theta(x))$

而 $KL(q\|P_\theta) = E_{q(x)}[\underbrace{\log q(x)}_{\text{与}\theta\text{无关!}} - \log P_\theta(x)]$

等价于 minimize $E_{q(x)}[-\log P_\theta(x)]$

或 maximize $E_{q(x)}[\log P_\theta(x)]$

也可以从MLE角度理解: $\log$ Likelihood $= E_{q(x)}[\log P_\theta(x)]$

$$\text{而} \log P_\theta(x_0) = \log \int P_\theta(x_{0:T})\, dx_{1:T}$$

$$= \log \int \frac{P_\theta(x_{0:T})\, q(x_{1:T}|x_0)}{q(x_{1:T}|x_0)}\, dx_{1:T}$$

Jensen不等式

$$\geq \underbrace{E_{q(x_{1:T}|x_0)}\left[\log \frac{P_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\right]}_{\text{VLB/ELBO}}$$

则 maximize $E_{q(x)}[\log P_\theta(x)] \Rightarrow$ maximize ELBO

$$\Rightarrow \text{Loss} = -\text{ELBO} = E_{q(x_{1:T}|x_0)}\left[\log \frac{P_\theta(x_{0:T})}{q(x_{1:T}|x_0)}\right]$$

经过推导, $\text{Loss} \approx \sum_{t=2}^{T} L_{t-1} = \sum_{t=2}^{T} E_{q(x_t|x_0)}\left[D_{KL}(q(x_{t-1}|x_t, x_0)\|P_\theta(x_{t-1}|x_t))\right]$

minimize Loss 等价于 minimize $E_{q(x_t|x_0)}\left[D_{KL}(q(x_{t-1}|x_t, x_0)\|P_\theta(x_{t-1}|x_t))\right]$

这也解释了为什么前面要假设 $P_\theta(x_{t-1}|x_t)$ 服从 Gaussian

因为 $q(x_{t-1}|x_t, x_0)$ 就是 Gaussian!

⑥ $P_\theta(x_{t-1}|x_t)$ 的方差, DDPM做了简化, 为定值 $\sigma_t^2 I$ (σ_t^2 可取 $\beta_t/\tilde{\beta}_t$, 实际上对应上/下限,

由于 $q(x_{t-1}|x_t, x_0)$ 的方差是 $\tilde{\beta}_t$, 这里取 $\sigma_t^2 = \tilde{\beta}_t$ 见 IDDPM)

⑦ 推导得 $L_{t-1} = E_{q(x_t|x_0)}\left[\frac{1}{2\sigma_t^2}\|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2\right]$